Biceps Femoris Aponeurosis Size: A Potential Risk Factor for Strain Injury?

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ABSTRACT

EVANGELIDIS, P. E., G. J. MASSEY, M. T. G. PAIN, and J. P. FOLLAND. Biceps Femoris Aponeurosis Size: A Potential Risk Factor for Strain Injury? *Med. Sci. Sports Exerc.*, Vol. 47, No. 7, pp. 1383–1389, 2015. **Purpose**: A disproportionately small biceps femoris long head (BFlh) proximal aponeurosis has been suggested as a risk factor for hamstring strain injury by concentrating mechanical strain on the surrounding muscle tissue. However, the size of the BFlh aponeurosis relative to BFlh muscle size, or overall knee flexor strength, has not been investigated. This study aimed to examine the relationship of BFlh proximal aponeurosis area with muscle size (maximal anatomical cross-sectional area (ACSAmax)) and knee flexor strength (isometric and eccentric). **Methods**: Magnetic resonance images of the dominant thigh of 30 healthy young males were analyzed to measure BFlh proximal aponeurosis area and muscle ACSAmax. Participants performed maximum voluntary contractions to assess knee flexion maximal isometric and eccentric torque (at 50° s^{-1} and 350° s^{-1}). **Results**: BFlh proximal aponeurosis area varied considerably between participants (more than fourfold, range = 7.5–33.5 cm^2, mean = 20.4 ± 5.4 cm^2, coefficient of variation = 26.6%) and was not related to BFlh ACSAmax (r = 0.04, P = 0.83). Consequently, the aponeurosis/muscle area ratio (defined as BFlh proximal aponeurosis area divided by BFlh ACSAmax) exhibited sixfold variability, being 83% smaller in one individual than another (0.53 to 3.09, coefficient of variation = 32.5%). Moreover, aponeurosis size was not related to isometric (r = 0.28, P = 0.13) or eccentric knee flexion strength (r = 0.24, $P \geq 0.20$). **Conclusion**: BFlh proximal aponeurosis size exhibits high variability between healthy young men, and it was not related to BFlh muscle size or knee flexor strength. Individuals with a relatively small aponeurosis may be at increased risk of hamstring strain injury. **Key Words:** APONEUROSIS MORPHOLOGY, MUSCLE–TENDON UNIT SIZE, HAMSTRINGS, BICEPS FEMORIS LONG HEAD, KNEE FLEXION STRENGTH, MRI

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The susceptibility of the hamstrings to strain injuries is well documented with the majority of these injuries located near the proximal myotendinous junction of the biceps femoris long head (BFlh) muscle (17). Although some risk factors for hamstring strain injuries have been identified (e.g., previous strain injury, strength imbalances, and muscle fatigue) (12,16), whether the anatomical structure of the BFlh muscle–tendon unit (MTU), including the aponeurosis, might influence injury risk has received very little attention. Only recently has a disproportionately small BFlh proximal aponeurosis been suggested as a potential risk factor for hamstring strain injury, following two studies that calculated higher localized tissue strains for individuals with narrow proximal aponeurosis using computational modeling and dynamic magnetic resonance imaging (MRI) (6,13).

In a pennate muscle, the force from the muscle fibers is transmitted to the tendon primarily via the aponeurosis. It seems reasonable to assume that a bigger and stronger muscle would have a larger and stronger aponeurosis and tendon to effectively and safely transfer the contractile force to the bone. Therefore, a degree of scaling between the size of the force generator and force transmitters seems intuitive. Within the quadriceps, the vastus lateralis (VL) aponeurosis area has been found to be in proportion to total quadriceps volume (1). However, whether this is the case for the hamstrings remains largely unknown. A preliminary report suggested that the width of the BFlh proximal aponeurosis was highly variable between individuals and unrelated to the size of the BFlh muscle (8), suggesting that the force transmitter may not be proportional to the force generator. If this is the case, a disproportionately small BFlh proximal aponeurosis may concentrate mechanical strain on the surrounding muscle tissue (5,6,13) and be a risk factor for hamstring strain injury. However, in this preliminary report, aponeurosis width was measured at a single arbitrary point along the muscle, which may be a poor reflection of the size of the aponeurosis. In contrast, measuring the whole contact interface between the muscle and aponeurosis may better reflect the concentration of mechanical strain at this interface.

From a functional perspective, aponeurosis size might be expected to be most strongly related to the maximum force transmitted through these tissues and thus muscle strength. Although experimental measurement of the force-generating capacity of the BFlh may not be possible *in vivo*, knee flexor torque, which is primarily due to hamstring activation, can be assessed. Maximal isometric torque is a convenient measurement

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of muscle function, although higher torques can often be achieved eccentrically, which likely explains the high risk of BFlh MTU strains during eccentric actions (9). However, the relationship between aponeurosis size and isometric or eccentric muscle strength has yet to be investigated.

The aim of this study was to examine the relationship of BFlh proximal aponeurosis area with muscle size (ACSAmax and volume) and knee flexor strength (isometric and eccentric). Based on the role of the aponeurosis as a force transmitter within the MTU, we hypothesized that BFlh proximal aponeurosis area would be positively related to muscle size and strength.

METHODS

Participants

Thirty healthy recreationally active participants (age, 20.7 ± 2.6 yr; height, 1.79 ± 0.07 m; body mass, 72.2 ± 7.2 kg (mean ± SD)) took part in this study. Participants had a low to moderate level of physical activity and were not involved in systematic physical training or had any previous experience of strength/power training (i.e., weight training or plyometrics) of the lower body musculature. Their physical activity was assessed with the International Physical Activity Questionnaire short format (www.ipaq.ki.se/downloads.htm) (4), and their average energy expenditure was 1826 ± 936 MET·min·wk^{-1}. After completing the physical activity and health screen questionnaires, participants provided written informed consent for their participation in this study, which was approved by the Loughborough University Ethical Advisory Committee. All participants were healthy with no history of musculoskeletal problems or injuries of the lower back, pelvis, or legs. Participants were instructed not to take part in any unaccustomed or strenuous physical activity for at least 2 d before each laboratory visit.

Overview

All participants visited the laboratory on four separate occasions 7 d apart at a consistent time of the day (11:00–16:00 h) for measurements on the knee flexors of their dominant leg (defined as the preferred leg when kicking a ball). The first session involved collection of the anthropometric data and familiarization with the isometric and eccentric strength measurements. The second session involved the measurement of the knee flexion isometric strength and further familiarization with the eccentric strength measurements. In the third session, participants repeated the isometric strength testing and also performed the eccentric strength measurements. The final session involved MRI of the participants' thigh to assess the BFlh proximal aponeurosis area, hamstring maximal anatomical cross-sectional area (ACSAmax), and BFlh/semitendinosus (ST) conjoint proximal tendon cross-sectional area (CSA).

Measurements and Data Analysis

Isometric strength. Participants lay in a prone position on a custom-made isometric dynamometer with hip and knee joint angles of 40° and 30°, respectively (0° = full extension) (Fig. 1). These angles were selected because of their relevance to the angles during the late swing phase in sprinting (7) when hamstring strains are thought to occur. To minimize any extraneous movements, participants were fastened with two straps across the hips, a strap over the lower back and a strap over the distal thigh just above the knee joint. A metal ankle cuff with a lining of high-density neoprene was placed ~2 cm above the medial malleolus, and the distal leg was tightly secured to the cuff with straps. The distance between the knee joint space and the centre of the ankle cuff was measured and used for calculation of the knee flexion torque. Force was measured with a calibrated strain gauge (linear response up to 500 N, Force Logic UK, Swallowfield, United Kingdom) connected in series to the ankle cuff and positioned perpendicularly to the tibia. The force signal was amplified (×1000) with an analog-to-digital converter (Micro 1401-3; CED, Cambridge, UK) and sampled at 2000 Hz with a PC using Spike 2 software (CED). The force signal was filtered with a fourth-order Butterworth filter with a low-pass cut-off frequency of 500 Hz.

After a standardized warm-up of progressively harder contractions, participants performed 3 maximal voluntary contractions (MVC), in which they were instructed to flex the knee as "hard" as possible for 3–5 s with 30 s of rest between the contractions. A computer screen provided real time visual feedback by displaying the torque response. After the first MVC a target cursor was positioned at peak torque achieved so far and participants were encouraged to exceed this during subsequent attempts. Standardized verbal encouragement was given throughout the MVC. All isometric torque values were gravity corrected by subtracting resting torque from peak torque. Isometric strength was defined as the highest instantaneous torque during any of the MVC within that session. Data presented is an average of the two sessions. The repeatability of isometric strength measurements between the two testing sessions was high (coefficient of variation (CV) = 3.9%).

Eccentric strength. The participants were seated on the dynamometer chair (Con-Trex MJ; CMV AG, Dübendorf,

FIGURE 1—Isometric measurements of the knee flexors were made with this custom-made isometric dynamometer in the joint configuration shown. Eccentric torque measurements were made with a Con-Trex isokinetic dynamometer.

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Switzerland) with a hip angle of 60° (0° = full extension). This was the most reclined position that could be obtained without participants sliding forward during contractions. However, this hip angle is similar to that during late swing phase in sprinting (7). Two three-point belts secured the torso, and additional straps tightly secured the pelvis and the distal thigh of their dominant leg. A brace was also placed in front of the noninvolved leg. The alignment of the knee joint with the dynamometer rotational axis was performed during isometric contractions of >50% of isometric strength at a knee joint angle of ~65°. The dynamometer's shin brace was placed posterior to the shank ~2 cm above the medial malleolus before the shank was tightly secured to the dynamometer lever arm. The range of motion was established and anatomical zero was set at full extension of the knee joint. Passive torque measurements were recorded while the tested leg was passively moved through the full range of motion, and thereafter, active torque values were corrected for passive torque.

Participants performed a standardized warm-up protocol with five submaximal contractions of progressively higher intensity. After the warm-up, the knee flexors were tested using a protocol of concentric–eccentric contractions at $50°\cdot s^{-1}$ and $350°\cdot s^{-1}$ in this order. Participants performed two sets of two and two sets of three concentric–eccentric contractions at $50°\cdot s^{-1}$ and $350°\cdot s^{-1}$, respectively, over ~100° of range of motion. There was ≥1 min of rest between each set and ≥2 min of rest between velocities. Participants were instructed to grasp the handles next to the seat during maximal contractions. Standardized verbal encouragement was given by the same investigator, and online visual feedback of the crank torque was provided on a computer screen.

The torque, crank angle, and crank velocity signals were amplified (×1000) with an analog-to-digital converter (Micro 1401-3, CED) and sampled at 2000 Hz with a PC using Spike 2 software (CED). The data were smoothed at 15 Hz. The acceleration and deceleration phases were excluded to disregard torque overshoot during these phases (14), and the constant isovelocity period (within ±5% of the prescribed crank angular velocity) was identified. Finally, eccentric strength was defined as the highest instantaneous torque recorded within the isovelocity range of any eccentric contraction.

MRI. A 1.5-T MRI scanner (Signa HDxt, GE, Fairfield, CT) was used to scan the dominant leg in the supine position with the hip and knee joints extended. T1-weighted axial plane images were acquired from the anterior superior iliac spine to the knee joint space in two blocks, and oil-filled capsules were placed on the lateral side of the participants' thigh to help with the alignment of the blocks during analysis. The following imaging parameters were used: imaging matrix, 512 × 512; field of view, 260 mm × 260 mm; spatial resolution, 0.508 mm × 0.508 mm; slice thickness, 5 mm; and interslice gap, 0 mm. MR images were analyzed with Osirix software (version 4.0; Pixmeo, Geneva, Switzerland).

BFlh aponeurosis area was defined as the contact interface distance between the BFlh muscle and the proximal aponeurosis outlined in each image where the aponeurosis was identifiable, multiplied by the slice thickness (Fig. 2A). The contact interface distance in each slice included both the internal and the external aponeurosis. The BFlh aponeurosis/muscle area ratio was calculated by dividing the BFlh proximal aponeurosis area by the BFlh muscle ACSAmax (see below). To produce average muscle–aponeurosis contact interface distance data for the cohort, individual values were normalized to muscle length. This involved interpolation of individual muscle–aponeurosis contact interface data every 5% of muscle length. BFlh aponeurosis length was calculated as the sum of the slices where the aponeurosis was identifiable, multiplied by the slice thickness. For comparison with previously published data, BFlh aponeurosis width was measured according to the methods of Handsfield et al. (8), i.e., width of the aponeurosis in the most distal image at which the proximal aponeurosis was external of the BFlh.

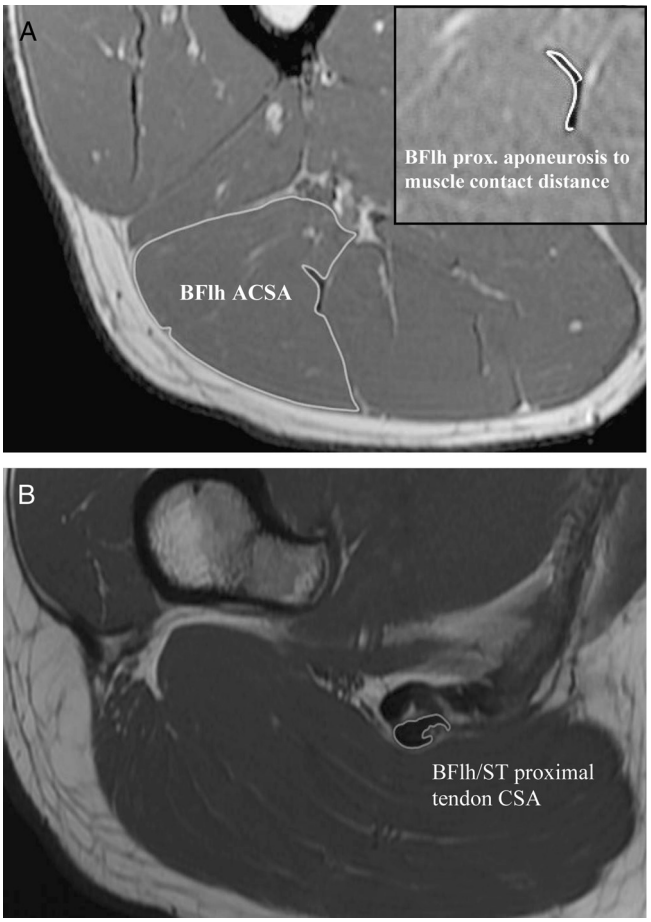

FIGURE 2—Example MR images of BFlh ACSA (main) and BFlh proximal aponeurosis to muscle contact distance (inset) (A) and measurement of the BFlh/ST proximal tendon CSA (B).

The BFlh, biceps femoris short head, ST, and semimembranosus muscles were manually outlined in every third image starting from the most proximal image in which the muscle appeared. The largest anatomical CSA of each muscle was defined as ACSAmax, and muscle volume was calculated using cubic spline interpolation (GraphPad Prism 6; GraphPad Software, Inc., La Jolla, CA). To validate the use of every third

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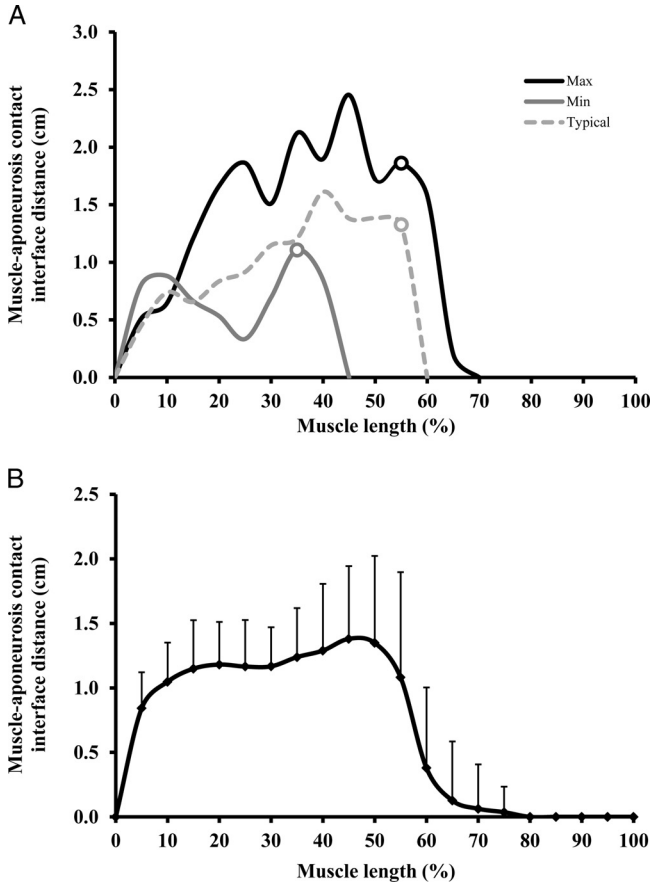

FIGURE 3—Muscle–aponeurosis contact interface distance along the length of the BFlh muscle (interpolated data every 5% of muscle length). A. Three individual participants (maximum, minimum, and typical (midrange) aponeurosis area). The circles indicate the relative muscle length where the aponeurosis width measurement was performed on each individual. B. Group mean ± SD.

image for the volume calculations, all images from six randomly selected participants were analyzed and the two methods (all images vs every third image) were compared. The average difference between methods was 1.52% (0.22 cm^2) for BFlh ACSAmax, 1.60% (0.20 cm^2) for total hamstring ACSAmax, 0.30% (0.66 cm^3) for BFlh volume, and 0.18% (1.52 cm^3) for total hamstring muscle volume. BFlh muscle length was calculated as the sum of all images where the muscle appeared multiplied by the slice thickness. BFlh/ST proximal tendon CSA was measured in the image immediately before the first image in which the ST muscle appeared (Fig. 2B). All manual segmentation measurements were completed by the same investigator. To examine reliability of the analysis procedures, the images from six randomly selected participants were reanalyzed a week later. The CV was on average 4.0% for the aponeurosis contact area, 0.6% for muscle volume, 1.1% for ACSAmax, and 5.5% for the BFlh/ST proximal tendon CSA.

Statistical Analysis

One participant did not complete the eccentric strength assessment. Data are presented as mean ± SD. Bivariate relationships were examined using Pearson product moment correlations between the dependent variables, and the level of significance was set at $P < 0.05$. All statistical procedures were performed with IBM SPSS 20 (IBM Corporation, Armonk, NY).

RESULTS

Descriptive data on size of the MTU components and knee flexor strength. BFlh proximal aponeurosis area varied considerably (higher than fourfold) between participants ranging from 7.5 to 33.5 cm^2 (20.4 ± 5.4 cm^2, CV = 26.6%). This was a reflection of the fact that aponeurosis length was variable (16.7 ± 2.8 cm, range = 10.5–22 cm or 43%–75% of muscle length), and muscle–aponeurosis contact interface distance was also variable along the aponeurosis length (Fig. 3).

Individual aponeurosis width measurements using a previously published method (8) appeared to occur at an arbitrary point along the aponeurosis (Fig. 3A, i.e., not at peak aponeurosis width or a consistent point along the aponeurosis). Aponeurosis width measured in this way was 0.43 ± 0.24 cm (range = 0.19–1.22 cm, CV = 56.4%).

Participants had a mean BFlh ACSAmax of 13.6 ± 2.2 cm^2 (CV = 16.2%), whereas their BFlh muscle volume was 214.7 ± 37.2 cm^3 (CV = 17.3%). BFlh muscle length was 29.3 ± 2.6 cm. Large interindividual variability was also found in proximal BFlh/ST tendon CSA (0.43 ± 0.14 cm^2, range = 0.25–0.91 cm^2, CV = 32.3%). In respect of the strength measurements, knee flexor isometric strength was 131.0 ± 19.9 N·m while eccentric strength was 134.3 ± 24.9 N·m at $50°·s^{-1}$ and 118.2 ± 21.6 N·m at $350°·s^{-1}$.

Relationships between the size of the different MTU components. BFlh proximal aponeurosis area was not related to BFlh ACSAmax ($r = 0.04$, $P = 0.830$; Fig. 4) or volume ($r = 0.35$, $P = 0.055$). Consequently, the aponeurosis/muscle area ratio also exhibited high variability (sixfold), being 83% smaller in one individual than another

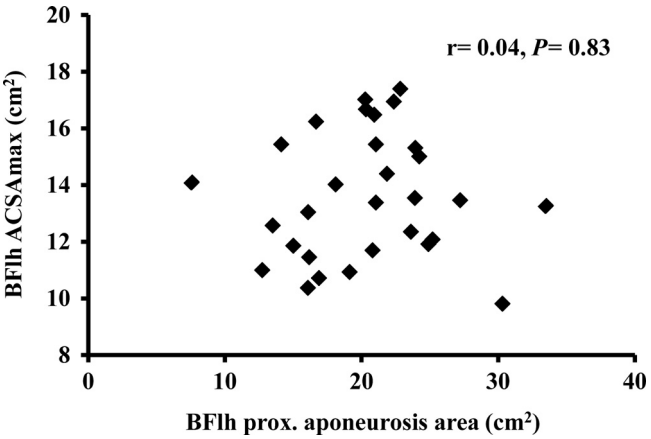

FIGURE 4—A scatter plot of BFlh proximal aponeurosis area and BFlh ACSAmax ($n = 30$). The individuals with the lowest and highest aponeurosis size (7.5 vs 33.5 cm^2, more than fourfold difference) had very similar-sized BFlh muscles (ACSAmax, 14.1 vs 13.3 cm^2) and thus aponeurosis/muscle size ratios of 0.53 versus 2.52.

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(range, 0.53 to 3.09; CV = 32.5%). BFlh proximal aponeurosis area presented a weak correlation with proximal BFlh/ST tendon CSA (r = 0.36, P = 0.049).

Relationships between the size of the MTU components with knee flexor strength. Although isometric strength was related to BFlh muscle ACSAmax and tendon CSA, there was no relationship with aponeurosis area (Fig. 5). Eccentric strength at both slow and fast velocities was related to BFlh muscle ACSAmax but not to aponeurosis area or tendon CSA (Table 1). Finally, overall hamstring ACSAmax was related to isometric strength as well as to eccentric strength at 50°·s^{-1} and 350°·s^{-1} (Table 1).

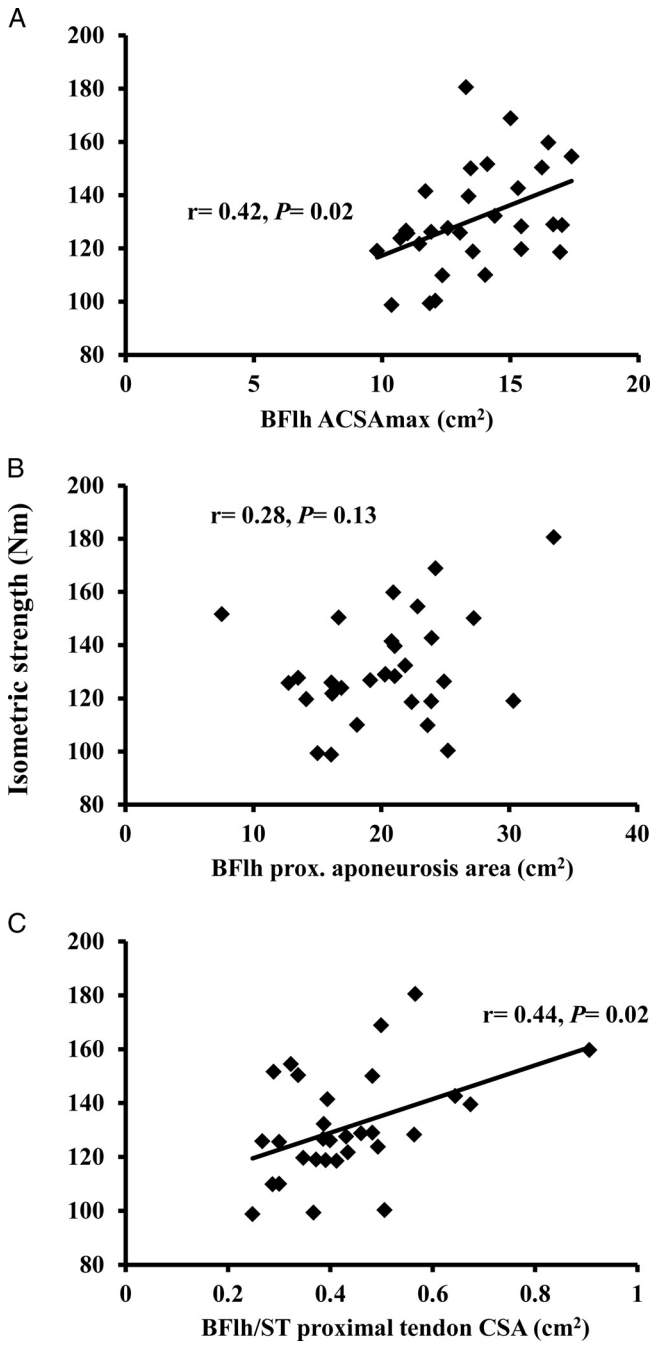

FIGURE 5—Scatter plots between knee flexor isometric strength and BFlh ACSAmax (A), BFlh proximal aponeurosis area (B), and BFlh/ST proximal tendon CSA (n = 30) (C).

TABLE 1. Bivariate correlations (r values) between the size of the hamstring muscle group and different components of the BFlh MTU with knee flexor isometric (n = 30) and eccentric (n = 29) strength.

		Hamstrings	BFlh		
		Muscle ACSAmax	Muscle ACSAmax	Aponeurosis area	Tendon CSA
Isometric strength		0.45*	0.42*	0.28	0.44*
Eccentric strength	50°·s^{-1}	0.63**	0.44*	0.24	0.29
	350°·s^{-1}	0.56**	0.50**	0.24	0.13

*P < 0.05.
**P < 0.01.

DISCUSSION

The present study examined the variability of the BFlh proximal aponeurosis size in healthy recreationally active population and the relationships of the aponeurosis size with BFlh muscle size and knee flexor function. The main finding was that the proximal aponeurosis size was highly variable between individuals, and in contrast to our hypothesis, it was not related to muscle size or knee flexor maximal isometric or eccentric strength. The disproportion between aponeurosis size and muscle size/strength suggests that individuals with a relatively small aponeurosis will be subject to greater mechanical strain in the muscle tissue surrounding the aponeurosis, which may predispose them to hamstring strain injuries.

Despite the homogenous nature of the recruited cohort, there was a large 4.5-fold variability in BFlh proximal aponeurosis area between participants that was substantially greater than the variability in BFlh muscle ACSAmax (1.8-fold). Moreover, contrary to our hypothesis, these two variables were unrelated, and consequently, the aponeurosis/muscle area ratio in this study ranged from 0.53 to 3.09 exhibiting a sixfold range, and being 83% smaller in one individual than another even within this relatively homogenous cohort. Interestingly, the individuals with the lowest and highest aponeurosis area in this study (7.5 and 33.5 cm^2 respectively) had similar BFlh ACSAmax (14.1 and 13.3 cm^2, Fig. 4), and consequently, their aponeurosis/muscle area ratios were 0.53 and 2.52. Similarly, from a functional perspective, our results showed that aponeurosis size was unrelated to knee flexor strength, whereas the sizes of the other components of the MTU (muscle and tendon area) were associated with muscle strength (Fig. 5).

The lack of relationship between aponeurosis size and muscle size may have important implications for the mechanical strain within the muscle tissue surrounding the aponeurosis. Based on modeling and *in vivo* measurements of mechanical strain, individuals with a relatively small aponeurosis/muscle size ratio would be expected to experience greater mechanical strain in the muscle tissue adjacent to the aponeurosis with a greater potential for injurious muscle strains (5,6,13). Therefore, our results combined with the fact that hamstring strain injuries typically occur near the BFlh proximal myotendinous junction (11) suggest that individuals with a low aponeurosis/

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muscle size ratio may be at an increased risk of hamstring strain injury.

This study is the first to directly examine the relationship between BFlh muscle and aponeurosis size. The only similar data we are aware of examined a different muscle and found VL aponeurosis area in a cohort of elite weightlifters and recreationally active males to be strongly related to total quadriceps muscle volume ($R^2 = 0.85$) (1). Although the adaptation of tendon in response to resistance training has been shown with increases in size and stiffness (10,15), it remains unknown whether the aponeurosis shows similar plasticity. The greater VL aponeurosis area exhibited by the elite weightlifters compared to the recreationally active students in the study of Abe et al. (1) suggests that this difference may be an adaptive response to resistance training. If this were the case, BFlh aponeurosis size may be modifiable through training and this could reduce the risk of strain injury. However, the effects of resistance training on aponeurosis size have yet to be investigated directly.

For comparative reasons, we measured aponeurosis width replicating the methods of a preliminary report (8), and the range of values obtained here (0.19–1.22 cm) was similar to their results. However, this method involved the measurement of aponeurosis width at an arbitrary point, which corresponded to different relative positions along both the aponeurosis and the muscle for each individual (Fig. 3A). This measurement was not at the point of peak muscle–aponeurosis contact interface distance. These limitations in the width measurement do not allow for any valid comparison with aponeurosis area or examination of the differences between individuals. Furthermore, the individuals exhibiting the lowest and highest aponeurosis area (more than fourfold difference) both had midrange aponeurosis widths (0.32 vs 0.63 cm, less than twofold difference). Therefore, the aponeurosis width measurement appeared to be a limited reflection of aponeurosis size and interindividual variability in this study.

Both aponeurosis and free tendon are considered to have a high safety factor (i.e., the ratio of failure stress to peak operating stress) such that the aponeurosis and tendon are capable of accommodating a range of loads well beyond the normal functional range with no risk of injury to these structures (2,3). This may partly explain the lack of relationship between muscle strength and aponeurosis size. Nevertheless, strain injuries typically occur within the muscle tissue adjacent to the aponeurosis rather than within the aponeurosis. Therefore, although a small aponeurosis may have a sufficient safety factor to preclude aponeurosis injury, it could make the adjacent muscle tissue vulnerable to injury.

An interesting observation made during the analysis of the MR images was that the BFlh aponeurosis extends not only longitudinally along the side of the muscle belly but also transversely into the muscle (Fig. 2A), in agreement with a previous report (6). Anecdotally, the proportion of the internal aponeurosis to the total aponeurosis area between our participants appeared highly variable. However, it is currently unknown how this aponeurosis morphology affects force transmission and stress distribution, and further study is needed to elucidate its relationship with muscle size and strength.

Despite the large number of studies examining possible risk factors for strain injuries, it is still unclear how to identify individuals at high risk of strain injury, especially those with no history of injury. The emerging evidence that aponeurosis size may be a risk factor for such injuries has significant implications. Establishment of such an anatomical feature as a risk factor would greatly help to distinguish at-risk individuals before an injury occurs. For that reason, a prospective study investigating aponeurosis area relative to muscle size and strength and recording which athletes go on to experience a strain injury would provide valuable information. Also, the possible interaction of aponeurosis area with other established risk factors (e.g., previous strain injury and strength imbalances) should be considered.

Some limitations of this study have to be considered. First, the knee joint axis of rotation was assumed to be passing through the knee joint space, which was identified using superficial anatomy. It was also assumed that knee flexor strength measurement *in vivo* reflected the force-generating capacity of BFlh muscle and the forces transmitted by the proximal aponeurosis.

In conclusion, the present study showed that the BFlh proximal aponeurosis size exhibits high variability within a relatively homogenous cohort of healthy young men, and it was not related to muscle size or knee flexor strength. Therefore, individuals with a relatively small aponeurosis may be at increased risk of hamstring strain injury.

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The authors declare that they have no conflicts of interest.
The results of the present study do not constitute endorsement by the American College of Sports Medicine.

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